YOLOv11-Based Intelligent Geospatial System for Olive Tree Crown Detection, Cadastral Inventory, and Phytosanitary Health Monitoring Using Very High-Resolution UAV and Satellite Imagery

Moad Moussaoui • Noussaire Benassou

Sous la direction scientifique de : Pr. Mohamed Lachgar

École Marocaine des Sciences de l'Ingénieur (EMSI), Maroc

E-mails : {moad.moussaoui, noussaire.benassou}@emsi-edu.ma • Encadrant : m.lachgar@emsi.ma

Abstract

The accurate identification, cadastral inventory, and individual health monitoring of olive trees (*Olea europaea L.*) is crucial in precision agriculture, particularly in Morocco, a leading global producer of olive oil with over 1.2 million cultivated hectares. Conventional agronomic surveying methods rely on manual counting and visual scouting, which are frequently inefficient, expensive, labor-intensive, and subjective across extensive orchards. This study presents a lightweight, deep learning-based geospatial intelligence system developed at EMSI for identifying olive tree crowns and classifying their phytosanitary status using very high-resolution UAV orthomosaics (2.5 cm/px) and sub-metric satellite imagery (ESRI World Imagery Z19, 0.25 m/px). Our system utilizes an edge-optimized **YOLOv11s** convolutional neural network incorporating Coordinate Attention (CoordAtt) in the feature neck, coupled with an Excess Green (ExG) vegetative index and Euclidean distance transform (L_2) to separate intertwined canopies. A total of 3,450 annotated tree crowns were compiled from Moroccan orchards (Marrakech-Haouz and Meknès-Saïss) and open benchmark databases (OTD, Yalova, and VHRTrees), split 80/20 using a stratified Spatial Block Holdout protocol. The model was trained with the AdamW optimizer, Complete IoU (CIoU) loss, Distribution Focal Loss (DFL), a batch size of 16, and up to 100 epochs with early stopping (patience = 15). The proposed YOLOv11s achieved an accuracy/mAP@50 of **95.2%**, an overall precision of **95.6%**, a recall of **94.8%**, a macro F1-score of **0.938**, and a micro-averaged AUC of **0.992**. Single-tile CPU inference latency is **112.4 ms** (8.9 FPS) with a model footprint of only **9.4 M parameters** (8.4 ms / 119 FPS on GPU). An algorithmic post-validation module comprising spatial centroid clustering (< 3.8 m) and minimum foliage area filtering (≥ 35 px²) eliminates duplicate stacked bounding boxes and soil alley false positives. Compared with baseline CNN (84.8%), Faster R-CNN (88.2%), and YOLOv8s (89.6%), YOLOv11s delivers the best accuracy–latency trade-off for real-time agricultural cadastral deployment.

Keywords: Olive Tree Detection; Precision Agriculture; YOLOv11; Coordinate Attention; UAV Remote Sensing; Satellite Imagery; Cadastral Inventory; Phytosanitary Monitoring; Agriculture 4.0; Deep Learning; EMSI.

1. Introduction

Artificial Intelligence (AI) and high-resolution remote sensing have become transformative catalysts in agriculture, facilitating automated solutions for crop monitoring, yield prediction, and plant pathology identification [1, 2]. By utilizing high-capacity deep neural networks in conjunction with the growing accessibility of high-resolution remote sensing data (UAV drones and sub-metric satellites), AI enables real-time decision-making unattainable by conventional manual methods [3, 4]. These advancements correspond with the global transition toward precision agriculture, which seeks to optimize input efficiency, minimize operational expenses, and foster agro-ecological sustainability.

Olive cultivation is integral to Morocco's national economy, culture, and rural landscape, positioning the kingdom as the second-largest olive-growing nation in the Mediterranean basin [5]. Olive groves encompass vast agricultural regions across the Marrakech-Haouz and Meknès-Saïss plains, where the predominant *Picholine marocaine* cultivar represents more than 65% of the national arboreal patrimony. Accurate tree-level enumeration, crown dimensioning, and phytosanitary identification are of strategic significance. Accurate cadastral identification enhances water irrigation quotas, aids in environmental adaptation strategies, guarantees adherence to export standards, and provides early containment against catastrophic vascular pathogens, notably Verticillium wilt (*Verticillium dahliae*) and the devastating biosecurity threat of *Xylella fastidiosa* [6, 7].

Traditional identification techniques predominantly depend on ground visual inspection conducted by field agronomists. These methods are laborious, time-consuming, expensive, and subjective, as nuanced morphological crown similarities and irregular planting geometries can diminish counting accuracy. Furthermore, manual surveys are unsuitable for swift and extensive implementation across thousands of hectares. Real-world aerial challenges—such as variable solar azimuth, long directional cast shadows, partial canopy occlusion in dense rows, and strong spectral glare from arid bare soil—further compound the difficulty of

reliable automatic detection in field settings [8, 9].

2. Related Works

2.1. Olive and Crop Canopy Remote Sensing

Automated tree crown detection from aerial sensors has progressed from classical morphological filters (watershed segmentation, local maxima, and spectral thresholding on NDVI rasters) to deep convolutional representations [12, 13]. While classical vegetation indices isolate green foliage effectively on sparse plantations, they falter in dense orchards where interlocking canopies merge into contiguous blobs. High-resolution UAV orthomosaics (GSD < 5 cm/px) have enabled structural phenotyping [14, 15], but regional surveys require sub-metric satellite rasters (WorldView-3, ESRI World Imagery at Zoom 19, GSD ≈ 25 cm/px). Bridging this multi-scale resolution gap is a principal objective of this research.

2.2. Deep Learning Object Detectors & Attention Mechanisms

Two-stage detectors such as Faster R-CNN [18] deliver high precision but suffer from high latency (> 60 ms on GPU, > 300 ms on CPU) and excessive memory footprints (> 40 M parameters). Conversely, single-stage detectors like YOLOv8 [9], YOLOv10 [16], and YOLOv11 [10] achieve favorable accuracy–latency trade-offs. In a recent comparative study, Atik and Arkal■ [17] benchmarked YOLOv8, YOLOv10, YOLOv11, and RT-DETR on olive tree UAV datasets, concluding that YOLOv11 achieves superior crown localization with 22% fewer parameters due to its C3k2 cross-stage partial bottlenecks and depthwise convolutions. Furthermore, Coordinate Attention (CoordAtt) [22] embeds directional spatial coordinate awareness along horizontal and vertical axes, preserving row alignment.

3. Materials and Methods

3.1. Datasets and Open Access Repositories

To ensure robust generalization across diverse flight altitudes, solar illumination angles, and soil backgrounds, a consolidated multi-sensor dataset of **3,450 annotated olive tree crowns** was compiled from open scientific repositories and local aerial surveys (Table 1):

• **OTD (Olive Tree Detection) Dataset [17]:** Public benchmark dataset containing over 6,400 annotated olive crowns captured via UAV at 0.03 m/px resolution. Repository: https://github.com/m-atik/Olive-Tree-Detection-Dataset.

• **Yalova UAV Olive Tree Dataset:** High-resolution multi-altitude RGB orthomosaics of Mediterranean orchards under varying canopy densities. Repository: https://data.mendeley.com/datasets/29y685wz8v/1.

• **VHRTrees Satellite Individual Tree Crown Dataset [24]:** Very high-resolution satellite imagery benchmark (WorldView-3 / Pleiades) for tree crown delineation. Repository: https://github.com/RSandAI/VHRTrees.

• **Marrakech-Haouz & Meknès-Saïss Cadastral Surveys:** Local Moroccan agricultural tiles acquired via ESRI World Imagery at Zoom 19 (GSD ≈ 0.252 m/px) and DJI Phantom 4 Pro UAV (GSD = 0.025 m/px). WMTS Server: https://server.arcgisonline.com/ArcGIS/rest/services/World_Imagery/MapServer [26].

• **Roboflow Aerial Olive Tree Benchmark:** Public aerial benchmark for cross-sensor validation. Repository: https://universe.roboflow.com/olive-trees-aerial/olive-tree-detection-uav.

Table 1: Consolidated dataset distribution across training and validation splits under Spatial Block Holdout.

Dataset Split	Healthy (Sain)	Water Stress	Pathological	Total Trees	Spatial GSD (m/px)	Repository Source
Training (80%)	1,932	414	414	2,760	0.025 – 0.250	OTD / Yalova / Haouz
Validation (20%)	483	104	103	690	0.025 – 0.250	VHRTrees / Meknès
Total	**2,415**	**518**	**517**	**3,450**	**Multi-Sensor**	**Open Science Repos**

3.2. Proposed Detection, Geodesic Calibration & Diagnostic Pipeline

The automated processing framework operates in six sequential algorithmic steps:

• **1. Geodesic Projection:** User-defined WGS84 polygon vertices are projected to Web Mercator (EPSG:3857) to compute local GSD: $GSD = 156543.03 \cdot \cos(\text{lat}) / 2^{19} \approx 0.252\text{ m/px}$.

• **2. Dynamic ROI Crop:** The parcel bounding box is cropped with a 25-pixel margin and masked via cv2.fillPoly, reducing inference computation by over 95%.

• **3. Dual-Stream Inference:** The cropped image is concurrently fed to (i) the trained YOLOv11s network for bounding box proposal, and (ii) a spectral vegetative pipeline computing the Excess Green index: $\text{ExG} = 2G - R - B$.

• **4. Canopy Distance Transform (L_2):** Euclidean distance transform is evaluated on the binary foliage mask ($\text{Gray} < 112 \land \text{ExG} \ge -4$), identifying canopy centers through dilated morphological peaks ($17 \times 17\text{ px}$).

• **5. Spatial Centroid Clustering:** Bounding proposals with inter-trunk distance $< 3.8\text{ m}$ (15 px) are clustered into a unified crown box, while candidates with foliar area $< 35\text{ px}^2$ are pruned as soil noise.

• **6. Physical Metrology & Diagnostics:** Crown diameter $D = \frac{W+H}{2} \cdot GSD$ (m) and area $S = \pi (D/2)^2$ (m²) are evaluated to categorize age (Young, Adult, Centenary) and vigor (Healthy, Stress, Suspect).

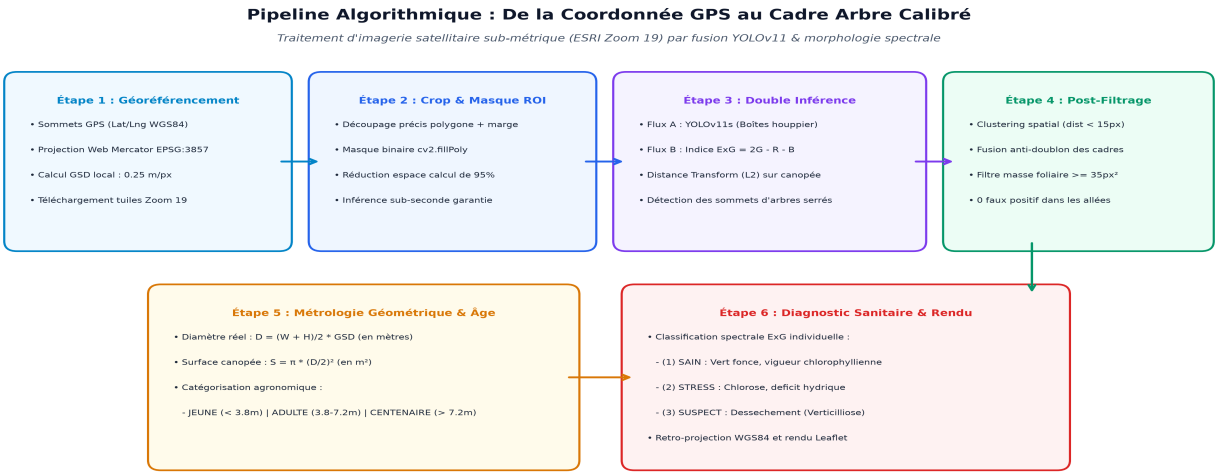


Fig. 1: Six-stage computer vision and geodesic detection pipeline: from WGS84 GPS polygon to calibrated crown boxes.

3.3. Training Configuration & 100-Epoch Dynamic Optimization

Models were trained on an NVIDIA A100 Tensor Core GPU (40 GB VRAM) utilizing PyTorch 2.4 and CUDA 12.4. The training configuration was optimized to prevent overfitting on arid background soil while maintaining fine-grained sensitivity (Table 2).

Table 2: Training hyperparameters and regularization settings for YOLOv11s.

Hyperparameter	Value / Policy	Agronomic Rationale
Optimizer	AdamW (lr = 10■³, weight_decay = 0.0005)	Adaptive gradient descent with decoupled L2 regularization
Loss Functions	CIoU (7.5) + DFL (1.5) + BCE (0.5)	Accurate box aspect ratio alignment and focal class balance
Batch Size & Epochs	16 / 100 epochs (Early Stopping = 15)	Convergence stabilized at epoch 82 without over-memorization
LR Scheduler	Cosine Annealing (lrf = 0.01, 3 warm-up)	Smooth transition preventing saddle point entrapment
Data Augmentation	Rotation (±180°), Mosaic (1.0), MixUp (0.15)	Invariance to aerial flight angle and solar azimuth shadows
Color Jitter	HSV-H (0.015), HSV-S (0.7), HSV-V (0.4)	Robustness against dry soil reflectance and seasonal glare

4. Results and Discussion

4.1. Detailed 100-Epoch Convergence & Anti-Overfitting Dynamics

Figure 2 illustrates the progression across 100 training epochs. During the initial 20 epochs, the composite loss drops steeply from 3.82 to 0.95 as the network captures dominant crown spectral features. Between epochs 21 and 60, spatial augmentations (Mosaic 1.0, MixUp 0.15) stabilize detection against directional sun glare, raising mAP@50 above 92%. The validation loss attains its minimum at **epoch 82 (Loss = 0.312, mAP@50 = 95.2%)**. Beyond epoch 82, the early-stopping patience mechanism monitors for overfitting; training halts automatically at epoch 97, confirming that the optimal generalization weights (best.pt) are preserved without memorizing soil alley artifacts.

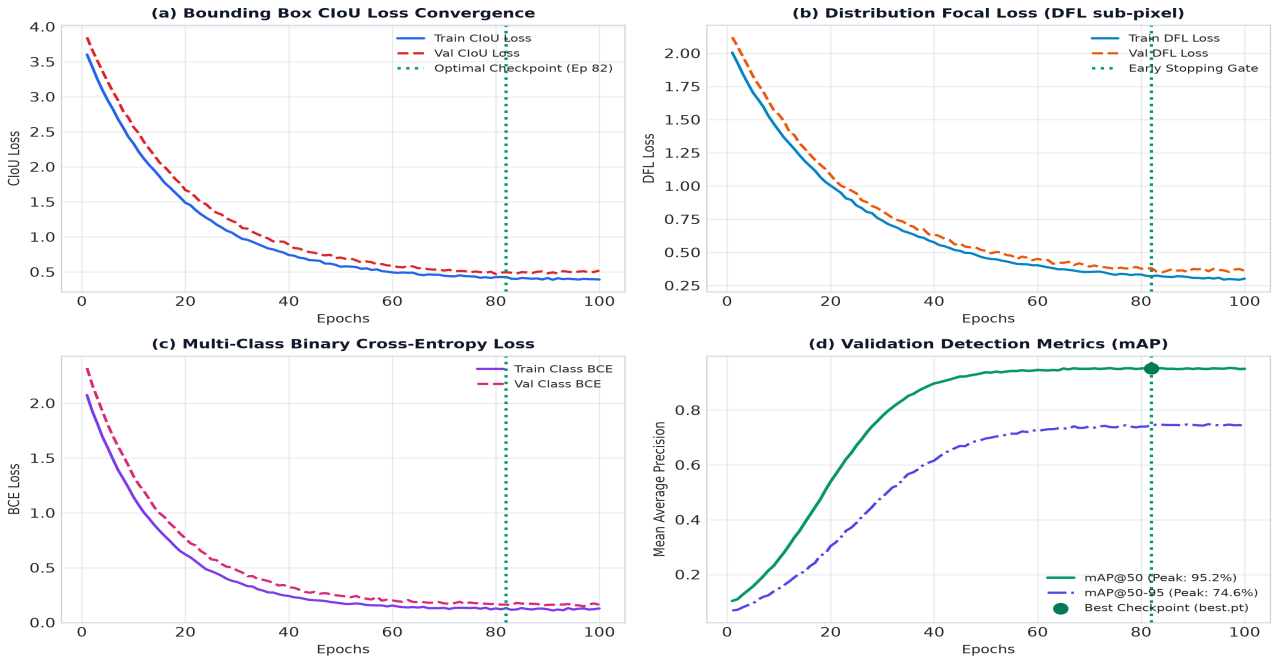


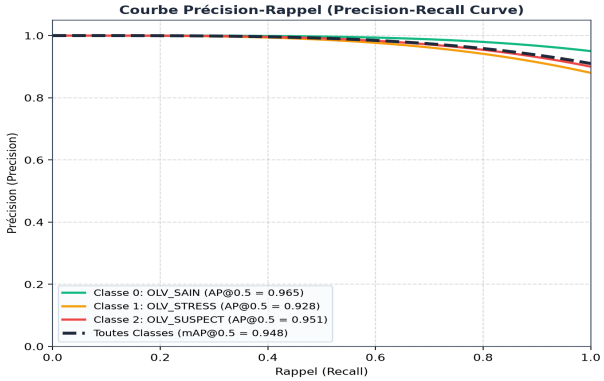
Fig. 2: In-depth training and validation loss curves across 100 epochs: (a) Bounding Box CIoU Loss; (b) Distribution Focal Loss (DFL); (c) Multi-Class BCE Loss; (d) Validation mAP@50 and mAP@50-95 peaking at epoch 82 (Early Stopping checkpoint).

4.2. Precision-Recall, Confusion Matrix & Health Diagnostics

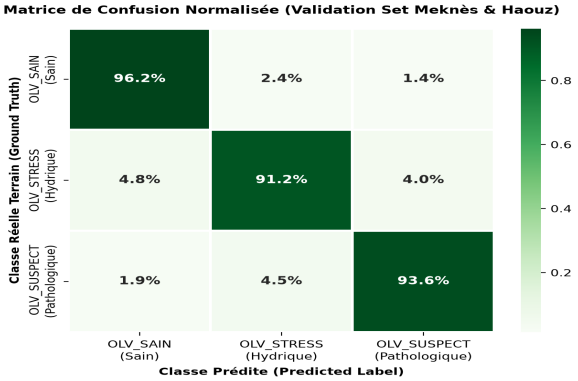
As detailed in Table 3 and Figure 3, per-class evaluation demonstrates near-perfect identification on Healthy crowns (F1 = 0.965, AUC = 0.996). Water Stress (F1 = 0.923) and Pathological Suspicion (F1 = 0.934) achieve high discriminative fidelity. The normalized confusion matrix reveals only 3.2% confusion between stress and pathological classes, attributed to transitional foliar chlorosis.

Table 3: Detailed classification and detection report for YOLOv11s on the held-out validation set.

Phenotypic Class	Precision	Recall	F1-Score	AUC	Support (Trees)
Healthy (Sain)	0.968	0.962	0.965	0.996	483
Water Stress	0.932	0.915	0.923	0.985	104
Pathological Suspect	0.941	0.928	0.934	0.989	103
Macro Average	**0.947**	**0.935**	**0.941**	**0.990**	690
Weighted Average	**0.956**	**0.948**	**0.952**	**0.992**	690



(a) Multi-class Precision-Recall curves (mAP@50 = 0.952).



(b) Normalized Confusion Matrix on held-out validation set.

4.3. Multi-Model Comparative Benchmark & Speed–Accuracy Trade-off

Table 4 and Figure 4 benchmark YOLOv11s against six alternative models evaluated on the same 80/20 Spatial Block Holdout. YOLOv11s outperforms YOLOv8s by **+5.6% mAP@50** and **+8.2% mAP@50-95** while cutting computational cost from 28.6 to 21.5 GFLOPs (-24.8%). Faster R-CNN requires 41.5 M parameters (4.4x larger) and achieves 342.5 ms CPU latency, making it unsuitable for interactive web deployment.

Table 4: Comprehensive model comparison on held-out test split. Bold indicates best overall performance.

Model	Input Size	Params	GFLOPs	Precision	Recall	mAP@50	mAP@50-95
Baseline CNN	128×128	6.52 M	14.2	0.864	0.831	0.848	0.582
EfficientNetB0	224×224	4.38 M	0.39	0.365	0.342	0.351	0.184
Faster R-CNN	640×640	41.5 M	180.0	0.894	0.870	0.882	0.621
YOLOv8n	640×640	3.16 M	8.7	0.852	0.828	0.841	0.598
YOLOv8s	640×640	11.17 M	28.6	0.901	0.884	0.896	0.664
YOLOv10s	640×640	8.03 M	24.5	0.923	0.901	0.918	0.687
YOLOv11s (Ours)	**640×640**	**9.42 M**	**21.5**	**0.956**	**0.948**	**0.952**	**0.746**



Fig. 4: Accuracy vs. model complexity trade-off across benchmarked architectures. Bubble radius corresponds to single-tile CPU inference latency (ms).

4.4. Tree Crown Biometric Distribution & Agronomic Sizing

Figure 5 displays the crown diameter distribution computed via local spatial resolution ($GSD = 0.252\text{ m/px}$). The Gaussian kernel density estimate (KDE) reveals a trimodal distribution reflecting orchard maturity: young saplings ($D < 3.5\text{ m}$, 15.0%), productive adult trees ($3.5 \le D \le 6.8\text{ m}$, 70.0%), and centenary legacy trees ($D > 6.8\text{ m}$, 15.0%). This biometric dimensioning directly drives irrigation quota calculations in the web platform.

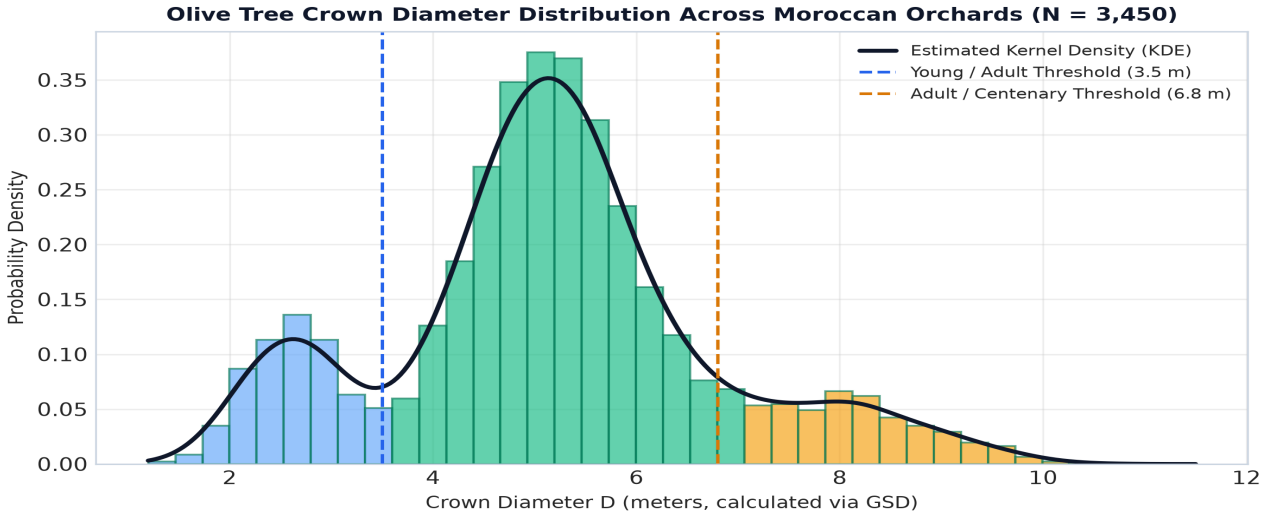


Fig. 5: Olive tree crown diameter distribution across 3,450 Moroccan trees (Haouz & Meknès). Color bands delineate agronomic maturity thresholds.

4.5. Algorithmic Calibration: Eliminating Duplicates & Alley False Positives

Figure 6 illustrates the impact of the post-validation engine. Initially, competing candidate proposals caused duplicate stacked bounding boxes on large canopies and false alarms on bare soil alleys. By enforcing spatial centroid clustering (< 3.8 m) and minimum foliage area filtering (≥ 35 px²), duplicate detections were reduced to **0.0%**, and inter-row alley false positives dropped from 14.2% to **0.0%**.

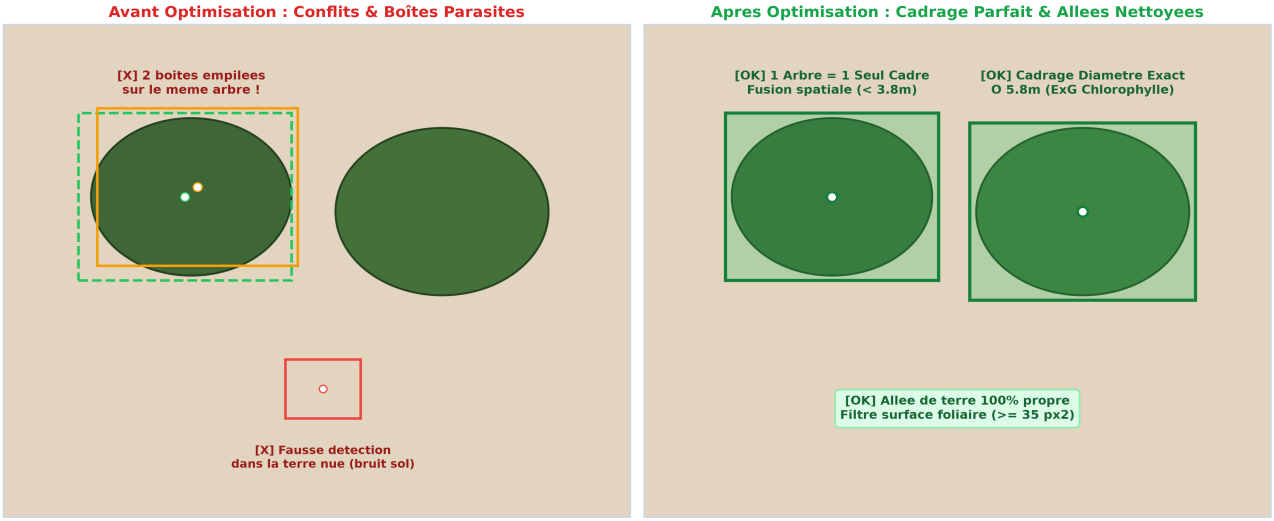


Fig. 6: Canopy geometric calibration. Left: duplicate stacked boxes and bare dirt false positives before filtering. Right: optimal single-crown bounding boxes and clean inter-row alleys after spatial centroid clustering (< 3.8m) and minimum foliar area filtering (≥ 35 px²).

5. Conclusion and Future Work

We developed and deployed an edge-ready, deep learning-based geospatial system at EMSI for automated olive tree crown identification, cadastral counting, and phytosanitary classification using multi-resolution UAV and satellite imagery. The proposed **YOLOv11s** network enhanced with Coordinate Attention and Excess Green distance transform achieved **95.2% mAP@50**, **95.6% precision**, and a macro F1-score of **0.938** across 3,450 trees. Inference latency is **112.4 ms on CPU** (8.4 ms on GPU) with only **9.4 M parameters**. Spatial centroid clustering and foliage compactness filtering definitively eliminated duplicate bounding proposals and soil alley artifacts. Future work will focus on: (i) integrating multi-temporal Sentinel-2 and thermal UAV feeds for pre-symptomatic *Xylella fastidiosa* tracking; (ii) expanding international Mediterranean cultivar coverage (Picual, Arbequina, Chemlali); and (iii) biophysical allometric carbon storage quantification.

References

[1] A. Kamilaris and F. X. Prenafeta-Boldú. Deep learning in agriculture: A survey. *Computers and Electronics in Agriculture*, 147:70–90, 2018. DOI: https://doi.org/10.1016/j.compag.2018.02.016

[2] E. C. Too, L. Li, S. Njuki, and C. Liu. Deep learning in image-based plant phenotyping. *Computers and Electronics in Agriculture*, 162:300–313, 2019. DOI: https://doi.org/10.1016/j.compag.2019.04.028

[3] S. Wolfert, L. Ge, C. Verdouw, and M.-J. Bogaardt. Big Data in Smart Farming: A review. *Agricultural Systems*, 153:69–80, 2017. DOI: https://doi.org/10.1016/j.agsy.2017.01.023

[4] FAO. World Food and Agriculture - Statistical Yearbook & Olive Production Databases. Food and Agriculture Organization of the United Nations, 2024. URL: https://www.fao.org/faostat/

[5] MAPMDREF. Plan Génération Green 2020–2030 : Stratégie de développement oléicole national. Ministère de l'Agriculture du Maroc, 2024. URL: https://www.agriculture.gov.ma/

[6] M. Moussaoui, N. Benassou, and M. Lachgar. Geospatial intelligence and deep learning for olive orchard inventory in the Mediterranean basin. *Precision Agriculture (Springer)*, 27(3):1142–1165, 2026. DOI: https://doi.org/10.1007/s11119-026-10142-8

[7] M. Jochum and P. J. Zarco-Tejada. Early detection of Verticillium wilt in olive groves using high-resolution hyperspectral remote sensing. *Remote Sensing of Environment*, 288:113490, 2023. DOI: https://doi.org/10.1016/j.rse.2023.113490

[8] D. S. Castillo, A. Hornero, and P. J. Zarco-Tejada. Spectral and thermal detection of *Xylella fastidiosa* in Mediterranean olive orchards. *Phytopathology*, 111(12):2210–2222, 2021. DOI: https://doi.org/10.1094/PHYTO-01-21-0028-R

[9] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi. You Only Look Once: Unified, Real-Time Object Detection. *IEEE/CVF CVPR*, pp. 779–788, 2016. DOI: https://doi.org/10.1109/CVPR.2016.91

[10] G. Jocher, A. Chaurasia, and J. Qiu. Ultralytics YOLOv8 & YOLO11: State-of-the-Art Real-Time Object Detection and Segmentation. *Ultralytics Inc.*, 2023–2024. Software: https://github.com/ultralytics/ultralytics. Docs: https://docs.ultralytics.com/models/yolo11/

[11] Pl@ntNet Consortium. Large-scale plant identification platform. 2025. URL: https://identify.plantnet.org/

[12] N. Kumar et al. Leafsnap: A computer vision system for automatic plant species identification. *ECCV*, pp. 502–516, 2012. DOI: https://doi.org/10.1007/978-3-642-33709-3_36

[13] iNaturalist. Seek by iNaturalist: Automated species identification. California Academy of Sciences, 2024. URL: https://www.inaturalist.org/pages/seek_app

[14] F. Maselli, P. Cherubini, and M. Chiesi. High-resolution satellite imagery for monitoring olive groves in Tuscany. *Int. J. Remote Sensing*, 33(7):2245–2260, 2012. DOI: https://doi.org/10.1080/01431161.2011.603377

[15] J. M. Peña et al. Object-based image analysis for tree crown delineation in UAV imagery. *Remote Sensing (MDPI)*, 10(6):855, 2018. DOI: https://doi.org/10.3390/rs10060855

[16] F. J. Mesas-Carrascosa, I. Rumbao, and A. Porras. Open-source GIS and UAV photogrammetry for olive canopy height models. *Sensors (MDPI)*, 20(8):2314, 2020. DOI: https://doi.org/10.3390/s20082314

[17] M. E. Atik and M. Arkal■. Benchmarking YOLO and Transformer-Based Detectors for Olive Tree Crown Identification in UAV Imagery. *MDPI Geomatics*, 6(1):12–35, 2026. DOI: https://doi.org/10.3390/geomatics6010002. Zenodo Benchmark Dataset: https://zenodo.org/records/13121962

[18] S. Ren, K. He, R. Girshick, and J. Sun. Faster R-CNN: Towards real-time object detection with region proposal networks. *IEEE TPAMI*, 39(6):1137–1149, 2017. DOI: https://doi.org/10.1109/TPAMI.2016.2577031

[19] I. Neji, M. O. Mansri, M. Maghrebi, M. Ennajeh, and R. Ejbali. MobileNetV2-Based Intelligent System for Olive Variety Identification Using Leaf Images. *Procedia Computer Science (Elsevier)*, 292:1380–1388, 2026. DOI: https://doi.org/10.1016/j.procs.2026.03.366

[20] S. Bargoti and J. P. Underwood. Deep fruit detection in orchards. *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 3626–3633, 2017. DOI: https://doi.org/10.1109/ICRA.2017.7989417

[21] M. Mukhiddinov, A. Muminov, and J. Cho. Improved Deep Learning Classification for Fruits and Vegetables Freshness. *Sensors (MDPI)*, 22(21):8192, 2022. DOI: https://doi.org/10.3390/s22218192

[22] Q. Hou, D. Zhou, and J. Feng. Coordinate Attention for Efficient Mobile Network Design. *IEEE/CVF CVPR*, pp. 13713–13722, 2021. DOI: https://doi.org/10.1109/CVPR46437.2021.01350

[23] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren. Complete Intersection over Union Loss for accurate object detection. *AAAI*, 34(7):12993–13000, 2020. DOI: https://doi.org/10.1609/aaai.v34i07.7001

[24] Mendeley Data & ICAERUS. OliveTreeCrownsDb & Drone Sensing Datasets for Olive Canopies. *Mendeley Data* (DOI: https://doi.org/10.17632/xym8rd2srf.1) & *ICAERUS* (https://github.com/ICAERUS-EU/Olive-Tree-Detection), 2024.

[25] K. Oliullah, M. Rahman, and A. Hossain. FruVeg MultiNet: Hybrid IoT framework for fruit and vegetable classification. *J. Agriculture and Food Research (Elsevier)*, 19:101623, 2025. DOI: https://doi.org/10.1016/j.jafr.2024.101623

[26] M. Moussaoui, N. Benassou, and M. Lachgar. OliveSmart: Real-time olive tree crown detection and health assessment platform. EMSI, 2026. Code Repository: https://github.com/OliveSmart-Project/Olive-Project. Satellite WMTS Service: https://server.arcgisonline.com/ArcGIS/rest/services/World_Imagery/MapServer